Majid Babaei, Ph.D.

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Summary

I am a tenure-track Assistant Professor in the School of Computing at the University of the Fraser Valley, working where model-driven engineering, software engineering, and artificial intelligence meet. My research aims to make AI-assisted software engineering reliable and its results checkable; it has appeared in venues such as ICSE, MODELS, and Software and Systems Modeling, received an ACM SIGSOFT Distinguished Paper Award, and is accompanied by open research software. I teach software engineering and programming, and have taught and coordinated programs at UFV, McGill, Concordia, and Queen's while co-advising graduate and undergraduate students. I serve the research community through conference organizing and program committee roles, and UFV as Chair of Industry Partnerships, drawing on earlier industry experience as a software developer and DevOps engineer.

Academic Appointments

Aug 2025–present
Assistant Professor (Tenure-Track)
University of the Fraser Valley School of Computing
Chair of Industry Partnerships.

Nov 2024–present
Associate Member (Adjunct Professorship is under review)
McGill University Electrical and Computer Engineering

Jan 2024–Jul 2025
Assistant Professor (Professional)
McGill University School of Continuing Studies
Academic Program Coordinator, Certificate in Computers and Information Technology.

Jan 2023–Jan 2024
Faculty Lecturer
McGill University School of Continuing Studies
Academic Program Coordinator, Micro-credential in Digital Product Management and Emerging Technologies.

2022–2023
Postdoctoral Research Fellow
Concordia University
Department of Computer Science and Software Engineering
Software robustness analysis. Supervisor: Prof. Yann-Gaël Guéhéneuc.

2022–2023
Visiting Researcher
McGill University Electrical and Computer Engineering
Knowledge-graph explainability for information retrieval. Host: Prof. Dániel Varró; industry project with Aggregate Intelligence.

Education

2018–Dec 2021
Ph.D. in Software Engineering
Queen's University, Kingston, Ontario
Thesis: *Regression Testing of Distributed Real-Time Embedded Systems in the Context of Model-Driven Development*.
Supervisor: Prof. Juergen Dingel.

2015–Nov 2017
M.Sc. in Automotive Software Engineering
Technische Universität Chemnitz, Germany
Thesis: *Conception and implementation of a pattern recognition system for analyzing Big Data from distributed embedded AMRA boxes*.
Supervisor: Prof. Martin Gaedke.

2007–May 2012
B.Eng. in Computer Engineering
Shahrood University of Technology, Iran
Supervisor: Ali Dostmohammadi.

Awards and Recognition

- **ACM SIGSOFT Distinguished Paper Award** 🏆, MODELS 2021.
- **Artifacts Available and Evaluated–Reusable badges** 📦✅♻️, MODELS 2020 research paper.

Publications

Refereed journal articles published or accepted

[J1] **M. Babaei**, N. Ezzati-Jivan, F. Hendijani Fard, and A. Hamou-Lhadj. RapidPay: Identifying and prioritizing self-admitted technical debt chains for effective debt management in software systems. *Empirical Software Engineering*, 2026. **To appear**.

[J2] **M. Babaei** and Y.-G. Guéhéneuc. RATEN: An efficient robustness analysis and test enhancement framework for state machines. *Software and Systems Modeling*, 2026.

[J3] **M. Babaei** and J. Dingel. Efficient regression testing of distributed real-time reactive systems in the context of model-driven development. *Software and Systems Modeling*, 2023, vol. 22, no. 5, pp. 1565–1587. doi: 10.1007/s10270-023-01086-5.

[J4] **M. Babaei**. A novel text and image encryption method based on chaos theory and DNA computing. *Natural Computing*, 2013, vol. 12, no. 1, pp. 101–107.

Refereed conference and workshop papers

[C1] **M. Babaei** and A. Heydarnoori. SkillSpec: A formally verifiable specification format for composing agent skills. *IEEE/ACM International Conference on Software Engineering (ICSE)*, Research Track, 2027. **To appear**.

[C2] A. Ezaz, G. Khodabandeh, **M. Babaei**, and N. Ezzati-Jivan. TAAF: A trace abstraction and analysis framework synergizing knowledge graphs and LLMs. *IEEE/ACM International Conference on Software Engineering (ICSE)*, 2026. doi: 10.1145/3744916.3787832.

[C3] **M. Babaei**, N. Ezzati-Jivan, and L. Wei. SIKG: A semantic impact knowledge graph approach for adaptive test selection and prioritization. *IEEE/ACM International Conference on Software Engineering Companion (ICSE Companion)*, Poster Track, 2026. doi: 10.1145/3774748.3795655.

[C4] G. Khodabandeh, A. Ezaz, **M. Babaei**, and N. Ezzati-Jivan. Utilizing graph neural networks for effective link prediction in microservice architectures. *ACM/SPEC International Conference on Performance Engineering (ICPE)*, 2025, pp. 19–30.

[C5] P. Chakraborty, **M. Babaei**, L. Tahmooresnejad, and N. Ezzati-Jivan. MemAdapt: Adaptive memory usage monitoring using irregularly sampled data. *International Conference on Collaborative Advances in Software and Computing (CASCON)*, 2024.

[C6] **M. Babaei**. Communication semantics for IoT-aware business process management systems. *IEEE/ACM International Workshop on Software Engineering Research and Practices for the IoT (SERP4IoT)*, 2024.

[C7] **M. Babaei** and M. Ben-Attia. ra4xstate: An efficient quantitative robustness analysis approach for statecharts. *IEEE/ACM International Workshop on Software Engineering Research and Practices for the IoT (SERP4IoT)*, 2023, pp. 46–53.

[C8] **M. Babaei** and J. Dingel. Efficient replay-based regression testing for distributed reactive systems in the context of model-driven development. *ACM/IEEE International Conference on Model Driven Engineering Languages and Systems (MODELS)*, 2021, pp. 89–100. **ACM SIGSOFT Distinguished Paper Award** 🏆.

[C9] **M. Babaei** and J. Dingel. MRegTest: A replay-based regression testing tool for distributed UML-RT models. *ACM/IEEE International Conference on Model Driven Engineering Languages and Systems Companion (MODELS Companion)*, 2021, pp. 620–624.

[C10] **M. Babaei**, M. Bagherzadeh, and J. Dingel. Efficient reordering and replay of execution traces of distributed reactive systems in the context of model-driven development. *ACM/IEEE International Conference on Model Driven Engineering Languages and Systems (MODELS)*, 2020, pp. 285–296. **Artifacts Available and Evaluated–Reusable** 📦✅♻️.

Refereed conference and workshop papers (continued)

[C11] **M. Babaei**, M. Bagherzadeh, and J. Dingel. MReplayer: A trace replayer of distributed UML-RT models. *ACM/IEEE International Conference on Model Driven Engineering Languages and Systems Companion (MODELS Companion)*, 2020.

[C12] **M. Babaei**, K. Jahed, and J. Dingel. Solving the RoboSoccer challenge problem with UML-RT and Papyrus-RT. *ACM/IEEE International Conference on Model Driven Engineering Languages and Systems Companion (MODELS Companion)*, 2019.

Submitted Manuscripts and Current Research

Submitted, under review, or revision requested

[S1] G. Khodabandeh, A. Ezaz, **M. Babaei**, and N. Ezzati-Jivan. Temporal Graph Models for Predictive Monitoring in Microservices. *Empirical Software Engineering*, 2026. **Revision requested**.

[S2] Z. Zhou, R. Chen, N. Yun, J. Chen, Y. Liu, **M. Babaei**, and L. Wei. Understanding and detecting login flow errors in Android applications (LIFTDroid). *IEEE Transactions on Software Engineering*, 2026. **Revision requested**.

[S3] **M. Babaei**, S. Farshidi, O. Adesina, N. Ezzati-Jivan, and L. Wei. AgentM2M: M2M transformations for agentic collaborations in software engineering. *IEEE/ACM International Conference on Software Engineering (ICSE)*, New Ideas and Emerging Results (NIER), 2027. **Submitted**.

[S4] S. Farshidi, **M. Babaei**, and B. Tekinerdogan. Mining design decisions and rationale evidence across research software projects. *International Conference on Mining Software Repositories (MSR)*, 2027. **Submitted**.

[S5] N. Ezzati-Jivan, M. Panahandeh, and **M. Babaei**. Was it my code? Evidence-based performance diagnosis for CI on shared GPUs. *IEEE/ACM International Conference on Software Engineering (ICSE)*, New Ideas and Emerging Results (NIER), 2027. **Submitted**.

[S6] **M. Babaei**. Trust Is Not Transitive: Verifying the Composition of LLM Guardrail Hierarchies. *IEEE Transactions on Software Engineering*, 2026. **Submitted**.

[S7] **M. Babaei**. Agent Slicing: Localizing Blind Spots in Multi-Agent AI Teams. *ACM Transactions on Software Engineering and Methodology (TOSEM)*, 2026. **Submitted**.

[S8] **M. Babaei**. Does Forgetting Help? Studying Context Reset Policies in LLM Fault Localization (MORPH-FL). *Journal of Systems and Software*, 2026. **Submitted**.

[S9] **M. Babaei**. Safe Coordination of Untrusted AI Agents. *IEEE Transactions on Software Engineering*, 2026. **Submitted**.

[S10] **M. Babaei**. use-hpc: Agent-configured, statically checked Slurm pipelines for large-scale GPU experiments. *Automated Software Engineering*, 2026. **Submitted**.

On-going projects

[S11] **M. Babaei**. RES-AI: Evidence-bound readiness for AI software engineering experiments on shared HPC infrastructure. **On-going**.

[S12] **M. Babaei**, S. Farshadi, N. Ezzati-Jivan, J. C. Davis, and L. Wei. When the team writes itself: Reliable and traceable LLM-assembled agent teams through checked model transformations (AutoM2M). **On-going**.

[S13] **M. Babaei**. Bad Habits Die Hard: A Model-Driven Approach to Traceable Self-Evolving Agent Harnesses (Heartness). **On-going**.

Research Software and Artifacts

Year	Software or artifact	Purpose and contribution	GitHub repository
2027	**SkillSpec**	Verifiable agent-skill contracts and composition repair.	`ai4se4ai-lab/swe-multiskillsbench`
2027	**AgentM2M**	Typed agent hand-offs and persistent transformation traces.	`ai4se4ai-lab/AgentM2M`
2026	**RATEN / ra4xstate**	State-machine robustness analysis and test enhancement.	`ai4se4ai-lab/RATEN-ra4xstate`
2026	**TAAF**	Trace knowledge-graph analysis and question benchmark.	`Alireza-Ezaz/TAAF`
2026	**SIKG**	Semantic impact analysis for test selection and prioritization.	`ai4se4ai-lab/SIKG`
2026	**RapidPay**	Technical-debt chains and SATD Impact Ripple ranking.	`ai4se4ai-lab/RapidPay`
2026*	**use-hpc**	Checked Slurm experiment graphs, runtime guards, and audits.	`ai4se4ai-lab/use-hpc`
2026*	**MORPH-FL**	Python fault localization with separate observation, hypothesis, and dialogue models for selective context resets.	`ai4se4ai-lab/MORPH-FL`
2026*	**Agent Slicing**	Pre-execution detection and localization of agent-team composition defects.	`ai4se4ai-lab/AgentSlicing`
2026*	**TINT**	Contract-based specification and compositional verification of LLM guardrail hierarchies.	`ai4se4ai-lab/trust-is-not-transitive`
2026*	**Safe Coordination**	Dependency-aware coordination of untrusted AI agent proposals.	`ai4se4ai-lab/safe-coordination-of-untrusted-ai-agents`
2021	**MRegTest**	Replay-based regression testing for distributed UML-RT models.	`MajidGitHubRepos/MRegTest`
2020	**MReplayer**	Deterministic execution-trace replay for distributed UML-RT models.	`MajidGitHubRepos/MReplayer`
2026*	**AutoM2M**	Checked agent-team composition, failure attribution, and repair.	`ai4se4ai-lab/M2MColab`
2026*	**RES-AI**	Readiness checks for coverage, freshness, and memory headroom.	`ai4se4ai-lab/RES-AI`
2026*	**Heartness**	Model-driven approach to traceable self-evolving agent harnesses.	`ai4se4ai-lab/heartness`

Years identify publication or target conference years; * marks current project activity (2026) where no publication year is assigned. Submitted work and prototypes are not publications.

Research Profile

From explicit models to checkable AI systems

My research asks how software can remain understandable, testable, and trustworthy as its behavior becomes more complex and increasingly shaped by AI. I connect **Model-Driven Engineering (MDE)**, **Software Engineering (SE)**, and **Artificial Intelligence (AI)** through three overlapping directions.

01 **Model behavior** — MDE

How can explicit abstractions make complex behavior testable?

My starting point was distributed real-time software: use behavioral models to isolate interactions, replay executions, and test the effects of change. This established a lasting principle: an explicit model makes assumptions visible and failures easier to explain.

Selected milestones: MReplayer [C11] (2020); MRegTest [C9] (2021); RATEN [J2] (2026).

02 **Learn from software evidence** — SE × AI

How can we recover useful structure from what systems actually do?

Operational systems expose detail that abstractions leave out. Traces, logs, and repositories are rich in evidence, yet difficult to interpret. I use knowledge graphs, semantic analysis, and AI—including LLMs—to turn this evidence into explanations and testing or maintenance decisions that developers can check.

Selected milestones: TAAF [C2] and SIKG [C3] (2026); RapidPay [J1] (2026, to appear).

03 **Engineer trustworthy agentic systems** — AI × MDE

How can stochastic components compose into dependable systems?

Agentic AI brings me back to explicit models. I use formalization and model checking to evaluate complementary aspects of AI agent teams: safe coordination, composition defects, and guardrail guarantees. Models, contracts, transformations, and provenance make team assumptions and interactions explicit and checkable.

Current work: Safe Coordination [S9] and TINT [S6] (submitted to IEEE TSE); Agent Slicing [S7] (submitted to ACM TOSEM); AgentM2M [S3] (submitted to ICSE 2027 NIER); AutoM2M [S12] (on-going).

This trajectory is cumulative: models help explain behavior; software evidence challenges and refines those models; and AI creates both new analytical capabilities and new coordination problems. The next step is to make the boundary between generative proposals and trusted system decisions precise: what may an agent assume, which evidence supports its actions, and which guarantees survive when components are combined or changed?

Explicit assumptions • Traceable evidence • Checkable guarantees

Teaching and Educational Leadership

University of the Fraser Valley

Term	Course	Title and sections	Enrolment
Fall 2025	COMP 370	Software Engineering; one section	26
	COMP 481	Functional and Logic Programming; one section	32
Winter 2026	COMP 370	Software Engineering; two sections	25 + 25
Fall 2026	COMP 370	Software Engineering; two sections	25 + 27
(current)	CIS 285	End User Training and Support; one section	23
	COMP 481	Functional and Logic Programming; one section	28
	COMP 155	Object-oriented Programming; one section	26

Total: nine sections, 237 course enrolments across completed and current terms (not unique students).

Earlier teaching and program coordination

- **McGill ECE**: ECSE 437, Software Delivery (Fall 2022 and Fall 2023); software-project instruction and structured peer feedback.
- **McGill SCS**: CCCS 300, Programming Techniques 1 (Winter 2024 and 2025); CCCS 301, Programming Techniques 2 (Fall 2023, Winter 2024 and 2025); CCCS 330, Database Design and Business Applications Development (Summer 2023); CCCS 610, Digital Thinking for Data Analysis (Fall 2024).
- **Program coordination, McGill SCS**: Certificate in Computers and Information Technology; Micro-credential in Digital Product Management and Emerging Technologies. Curriculum review and program redesign.
- **Concordia**: COMP 218, Fundamentals of Programming (Fall 2021 and Winter 2023).
- **Queen's**: ELEC 377, Operating Systems (Fall 2021); course-lecturer and teaching-assistant experience during the doctoral period.
- **St. Lawrence College**: applied AI and machine-learning teaching (2020–2022).
- Additional teaching areas across appointments include data-analytics programming with R and Python.

Student Supervision and Academic Advising

Current collaborative co-supervision: with Abbas Heydarnoori, two graduate students and one undergraduate on AI-powered fault localization and context engineering; with Fatemeh Hendijani Fard, graduate and undergraduate students on collaborative projects including *CodeFaithBench*, a new benchmark project on the faithfulness of LLM code explanations.

Completed master's theses co-advised

Brock University; academic advisor/co-advisor, with supervisor Prof. Naser Ezzati-Jivan.

- **Ghazal Khodabandeh** (Jan 2024–Sep 2025). *Link Prediction on Distributed Systems*. Publication: ICPE 2025. Subsequent position: Research Assistant, York University.
- **Alireza Ezaz** (Jan 2024–Sep 2025). *TAAF: A Trace Abstraction and Analysis Framework Synergizing Knowledge Graphs and LLMs*. Publication: ICSE 2026. Subsequent position: Data Quality Specialist, Cohere.

Completed undergraduate and project advising

- **Brock, Jan 2024–Apr 2025**: Bilal Bilal (semantic code coverage using topic models and LLMs); Yuvraj Sehgal (LLM-generated README files from source code).
- **McGill, Jan–Apr 2024**: Felicia Sun (review of AI-enabled continuous testing in DevOps); Zachary Hayden (LLMs and technical debt in DevOps).
- **McGill, Sep–Dec 2023**: Benakmoum Amine, Ammar Ryad, Zakaria Bouhemhem (DAPPER subscription application).
- **McGill, Sep–Dec 2022**: Michael Buchar, Ege Karadibak, Ezra Gomolin, Dominic Chan (decentralized NFT-locking application).

Research Funding and Proposal Development

Program or proposal	Role	Status	Project and partners
NSERC Discovery Grant (2026)	Sole applicant	Under review	*Checkable Abstractions for Trustworthy AI in Distributed Software Systems*. Five-year program on grounded diagnosis, calibrated forecasts, verified teams, and checked evaluation.
NSERC RTI	Applicant	Under review	*An Interconnected Computing Platform for Trustworthy Intelligent Systems*. Requested: CAD $41,680.
Amazon Research Awards (ARA)	Principal Investigator	Under review	*MAPLE: Model-Driven Engineering of Access Permissions for AI Agents*. Co-PI: Abbas Heydarnoori (Bowling Green State University). Requested: USD $70,000 cash and USD $18,200 in AWS Promotional Credits; proposed period Jun 2027–May 2028.
Mitacs Accelerate	Principal Investigator	Under review	Applied AI and educational technology with Enya-Learning; proposed research internships.
CIRA Net Good Grants	Applicant / UFV lead	Not funded	AI-assisted analysis of internet affordability and Indigenous-connectivity policies, with Caseway and Indigenous/community participants.
DND IDEaS Innovation Networks	UFV lead / initial recipient	Not funded	*ARMOR: Agnostic Real-time Monitoring and Operator Trust Repair Layer*. McGill and Caseway; verification, resilience testing, human–AI trust, and secure data fusion.
Community-based experiential learning	Contributor	Under review	UFV CECE and Ken Ryu Jujutsu Society; proposed student development of a mobile training application for Fall 2026.

Industry Partnerships and Knowledge Mobilization

Chair of Industry Partnerships, School of Computing, UFV (Fall 2025–present). Develop relationships supporting experiential learning, curriculum relevance, applied research, and professional pathways. Selected documented activities:

- **Tech Panel 2026**, Mar 12, 2026: organized with the Computing Student Association and support from colleagues and staff. Panel and networking session with Colin Schmidt (Zynim), Meer Anwar (PixelPaddle), Mason Paquette (Kwixand), and Stan Engelbrecht (ADAMnetworks).
- **Industry events, Fall 2025**: organized a cloud-computing talk by Craig Slack (Invero CEO and UFV alumnus), with staff support; co-managed a technical-interview event with Richard Clark (Crafty Penguins) and UFV career-education partners.
- **Curriculum integration, 2025–2026**: invited Nathan Leggatt (Intuit Mailchimp) into COMP 370; facilitated a Caseway engineer's presentation of Synthium requirements in Winter 2026.
- **External Program Advisory Committee**: chaired the Nov 27, 2025 meeting; advanced Terms of Reference updates, systematic industry feedback, alumni engagement, and discussion on bridging research and industry.
- **Exploratory collaboration**: AWS Canada Innovation Sandbox; HoloFi; CropVue Technologies; ROSC; and an Altifarm introduction connecting UFV software and hardware expertise. Discussions concerned research scope and student opportunities.
- **Partnership evaluation**: assessed EC-Council and Woddy opportunities; did not proceed where curriculum or grant fit was insufficient.
- **Community proposal development**: contributed to the Indigenous-connectivity policy proposal and the CECE/Ken Ryu experiential-learning proposal; proposed deliverables are distinct from completed outcomes.

Earlier research connections: Aggregate Intelligence (2022), explainable information retrieval, including mentoring three McGill ECE undergraduate researchers in Summer 2022; Humanitas (2022–2023), robustness analysis. Organizer, McGill SCS Tech Trends Speaker Series, Winter 2025.

Professional and Institutional Service

Conference organization

Year	Conference	Role	Official page
2026	**CASCON**	Publicity Co-Chair	conf.researchr.org/.../cascon-2026
2025	**ICSE**	Proceedings Co-Chair	conf.researchr.org/.../icse-2025
2025	**SEAMS**	Publicity Co-Chair	conf.researchr.org/.../seams-2025
2025	**SERP4IoT**	Publicity Chair	conf.researchr.org/.../icse-2025

Program committees

Year	Conference	Track	Official page
2026	**ASE**	Tools and Datasets	conf.researchr.org/.../ase-2026
2026	**FSE**	Student Research Competition	conf.researchr.org/.../fse-2026
2025	**ASE**	Tool Demonstrations	conf.researchr.org/.../ase-2025
2025	**MODELS**	Workshops	conf.researchr.org/.../models-2025
2025	**ICPE**	Data Challenge	icpe2025.spec.org
2024	**ASE**	Artifact Evaluation	conf.researchr.org/.../ase-2024

Institutional and professional committees

- **Chair of Industry Partnerships**, School of Computing, UFV (2025–present).
- **SCS Strategic Planning Committee**, McGill University (Sep 2023–Sep 2026).
- **Working Group on Revisions to the Guide to Reporting Alleged Violations of Academic Integrity**, McGill SCS (Jul 2023–Mar 2025).
- **Professional Development Committee**, Canadian Association for University Continuing Education (Apr 2023–Dec 2025).

Industry Experience

Nov 2021–Mar 2023 **Senior Software Developer and DevOps Engineer**
Imaginary Spaces, Canada
Row64 software development and debugging; CI/CD workflows for Pipeline-In-Box readers and writers.

May–Sep 2021 **Software Developer (Internship)**
National Research Council Canada, Ottawa
Representation learning for cybersecurity log data; advanced persistent threat detection for large-scale databases.

Jan–Dec 2017 **Software Developer**
Robert Bosch GmbH, Germany
Bosch IoT/AMRA datasets; React and React Native monitoring applications; log analysis for real-time embedded devices; customizable analysis workspaces.

Apr 2015–Dec 2016 **Software Developer**
EPAK GmbH, Leipzig, Germany
VSAT/TV antenna networking platform, simulation, web interfaces, and internal communication system.

May 2012–Apr 2015 **Software Developer**
Farazpajohan, Tehran, Iran
PKI-based digital-certificate development; Linux networking APIs; modular TCP/SSL server applications.

Additional Information

Interview with Majid Babaei: McGill School of Continuing Studies, Aug 21, 2023.
mcgill.ca/continuingstudies/article/interview-majid-babaei

Languages: English and Persian (reading, writing, speaking, understanding, and peer review).